

Week 09: Specifications and induction

Functional Programming in Lean

Learning goals

- ▶ Theorem statements as specifications
- ▶ Induction on a list
- ▶ Axiom audit

Core concepts

- ▶ Base case: empty list
- ▶ Cons case: head and induction hypothesis
- ▶ Goal simplification with definitions

Lean example

```
theorem append_nil (xs : List  $\alpha$ ) :  
  append xs [] = xs := by  
  induction xs with  
  | nil => rfl  
  | cons x xs ih => simp [append, ih]
```

Lean details

- ▶ `ih`: property of the tail
- ▶ `simp [append, ih]`: definition and hypothesis
- ▶ `#print axioms`: transitive axiom dependencies

Live coding

- ▶ Goals after induction xs
- ▶ Cons-case proof
- ▶ Axiom report

Check

Statement of ih in the cons case

Exercise

- ▶ `append [] ys = ys`
- ▶ Associativity of `append`